Investment Group Project Report

Group 6

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1 Introduction

This project aims to apply modern portfolio theory to achieve an efficient portfolio allocation. We evaluate the current investment policy of the GBF College Fund, which consists of 60% stocks and 40% bonds. We also explore the potential benefits of adding more asset classes to the portfolio.The data used in this study is sourced from CSMAR (https://data.csmar.com/) and WIND.

2 Current Investment Policy Evaluation

2.1 Performance Statistics

2.1.1 Calculation Process

Calculation of Portfolio Monthly Returns The portfolio monthly return for month i is calculated as the weighted sum of the individual asset returns for that month. The formula for the portfolio monthly return is:

$$r_{p,i} = \sum_{j=1}^{n} w_j \cdot r_{j,i}$$

where w_j is the weight of asset j in the portfolio, $r_{j,i}$ is the return of asset j for month i , and n is the total number of assets in the portfolio.

Calculation of Annualized Mean Returns The annualized mean return of a portfolio is calculated by taking the average of the monthly returns and then annualizing it. The formula for the annualized mean return is:

$$\text{Annualized Mean Return} = \left(\prod_{i=1}^{n}(1+r_i)\right)^{\frac{12}{n}} - 1$$

where r_i is the monthly return for month i .

Calculation of Portfolio Standard Deviation The portfolio standard deviation is a measure of the risk associated with the portfolio. It is calculated using the covariance matrix of the assets' returns and the weights of the assets in the portfolio. The formula for the portfolio standard deviation is:

$$\sigma_p = \sqrt{w^T \Sigma w}$$

where w is the vector of portfolio weights and Σ is the covariance matrix of the assets' returns.

Annualizing the Standard Deviation If the covariance matrix is based on monthly returns, the standard deviation calculated from it will also be a monthly standard deviation. To annualize it, we multiply by the square root of 12 (since there are 12 months in a year):

$$\sigma_{p,\text{annual}} = \sigma_p \times \sqrt{12}$$

Annualized Covariance Matrix The annualized covariance matrix used in the calculation is as follows:

	000012	000852	000905	399300	H11001
000012	9.4833e-05	-1.2080e-04	-1.5546e-04	-2.9315e-04	1.7144e-04
000852	-1.2080e-04	0.086738	0.071865	0.047346	-3.1362e-04
000905	-1.5546e-04	0.071865	0.063537	0.045378	-3.2626e-04
399300	-2.9315e-04	0.047346	0.045378	0.043806	-6.9087e-04
H11001	1.7144e-04	-3.1362e-04	-3.2626e-04	-6.9087e-04	4.5820e-04

Table 1: Annualized Covariance Matrix

2.1.2 Calculation Results

The performance statistics for the five assets and the portfolio are summarized in the table below:

Asset	**000012**	**000852**	**000905**	**399300**	**H11001**	**Portfolio**
Annualized Mean Return	4.33%	-0.12%	0.73%	1.08%	5.00%	3.11%
Standard Deviation	0.97%	29.45%	25.21%	20.93%	2.14%	14.43%
Sharpe Ratio	2.64	-0.06	-0.04	-0.03	1.51	0.09

Table 2: Performance Statistics of the Five Assets and Portfolio

2.1.3 Evaluation

The portfolio achieved a 3.11% annualized return with 14.43% standard deviation, yielding a Sharpe ratio of 0.094 at a 1.76% risk-free rate.Bond indices performed strongly with Sharpe ratios above 1.5, while equity indices, especially CSI 1000, underperformed with negative returns and low Sharpe ratios.This dragged down the portfolio's overall risk-adjusted return,indicating the current strategy inadequately compensates for the risk taken.

2.2 Portfolio Frontier

2.2.1 Critical Steps

The critical steps we use are listed below:

1. Data Preparation: Define the expected returns and risk (covariance matrix) for five assets.

2. Optimization Setup: Adjust the risk matrix to ensure stable calculations. Set rules: portfolio weights must add up to 100%, and returns must match target values.

3. Finding the Efficient Portfolio: For different target return, calculate the lowest possible risk using an optimization method.

4. Removing Invalid Calculations: Remove invalid or failed calculations.

2.2.2 Plot of the Efficient Frontier

We plot the portfolio frontier given only the five risky assets in which we are investing. Short sales are allowed when developing this frontier.

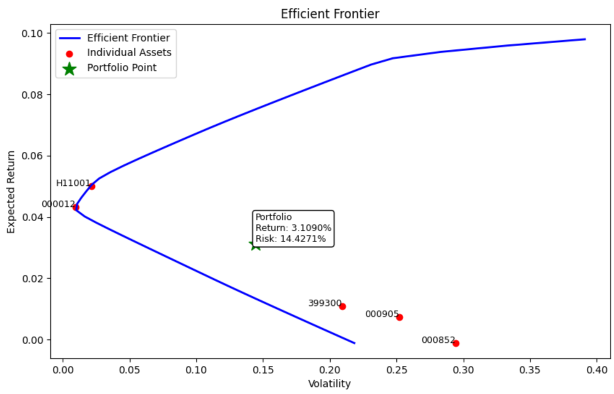

Figure 1: Plot of the efficient frontier and portfolio

2.3 Efficiency of the Current Portfolio

The green star portfolio,with 20% assigned equally to five assets, has an expected return of about 3.1% and a risk of about 14.4271% . It does not lie on the efficient frontier, indicating that there are other asset combinations that offer a higher return for the same risk or lower risk for the same return.

To find an efficient portfolio with the same expected return but lower risk, we would look for a point on the efficient frontier matching the expected return of the current portfolio. However, since the green star portfolio lies below the efficient frontier, no such point exists. Therefore, for question 3, we assume no efficient portfolio can deliver the same expected return as the current choice of risky portfolio that lies on the efficient frontier.

2.4 Optimal (Tangency) Portfolio

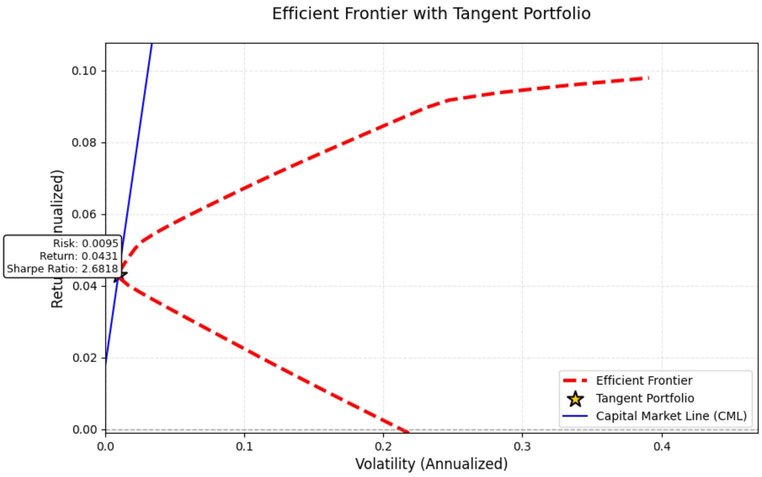

Figure 2: The Optimal (Tangency) Portfolio

By using Python, we plot the Capital Market Line (CML) and find the touching point, which is the portfolio with return 4.31%, risk 0.95%, and the Sharpe ratio 2.6818.

The concrete Python process is summarized as follows. The efficient frontier is then calculated by finding the minimum variance portfolios for different expected return levels using optimization techniques. A risk-free rate is set at 1.76% to establish the Capital Market Line (CML) and to calculate the Sharpe ratio. The optimal portfolio, known as the tangent portfolio, is determined by maximizing the Sharpe ratio, which identifies the portfolio that offers the best risk-adjusted return. Finally, the information about the weights of each asset in the tangent portfolio, along with the overall portfolio metrics, is presented and visualized through a table that includes the efficient frontier and the tangent portfolio.

From the result, we get the fact that the investment proportions required to achieve the optimal (tangency) portfolio with the five risky assets, given the possibility of investing at a risk-free rate, are shown in the below table.

Asset	000012	000852	000905	399300	H11001
Opt. Prop.:Optimal Proportion	99.39%	0.00%	0.00%	0.61%	0.00%

Table 3: Optimal Proportion of the Tangent Portfolio

3 Efficient Portfolio Analysis with Risk-Free Asset

3.1 Efficient Portfolio with Existing Portfolio's Expected Return

Assuming the ability to invest in a risk-free asset, the objective is to construct an efficient portfolio that delivers the same expected return as the original investment policy's portfolio, but with a lower standard deviation.

Part 2 shows that the current portfolio delivers an annualized rate of return around 3.11% and an annualized standard deviation around 14.43%. The risk-free rate is set at 1.76%, and the tangency portfolio achieves a return of 4.31% and standard deviation of 0.95%.

To obtain an efficient portfolio with an expected return equal to the current portfolio's annualized return, while allowing for investment in a risk-free asset, the following investment proportions were calculated:

- **Weight of Risk-Free Asset: 0.4712**
- **Weight of Tangency Portfolio: 0.5288**

The calculation of weights is enabled by Python, using the following formula:

$$w_{\text{tangency}} = \frac{R_{\text{original}} - R_{\text{risk-free}}}{R_{\text{tangency}} - R_{\text{risk-free}}}$$

This indicates an investment in both the risk-free asset and the tangency portfolio. The expected standard deviation of returns for this efficient portfolio is calculated to be 0.5092%, by the following formula:

$$\sigma_{\text{efficient}} = w_{\text{tangency}} \times \sigma_{\text{tangency}}$$

Specifically, the allocation of each asset to achieve the efficient portfolio that matches the return on the current portfolio is shown in the below table.

Asset	000012	000852	000905	399300	H11001	Risk-Free Asset
Opt. Prop.	52.5539%	0.0000%	0.0000%	0.3219%	0.0000%	47.1241%

Table 4: Optimal Proportion of the Efficient Portfolio Matching Existing Portfolio's Expected Return

The graphical representation in Figure 3 illustrates this point. The Efficient Portfolio on CML's positioned on the Capital Market Line, demonstrating its efficiency compared to the original portfolio by achieving the same return with lower volatility. The dashed black line represents the "Risk Difference," highlighting the reduction in risk achieved by moving from the original portfolio to the efficient portfolio on the CML, while maintaining the same expected return (3.11%).

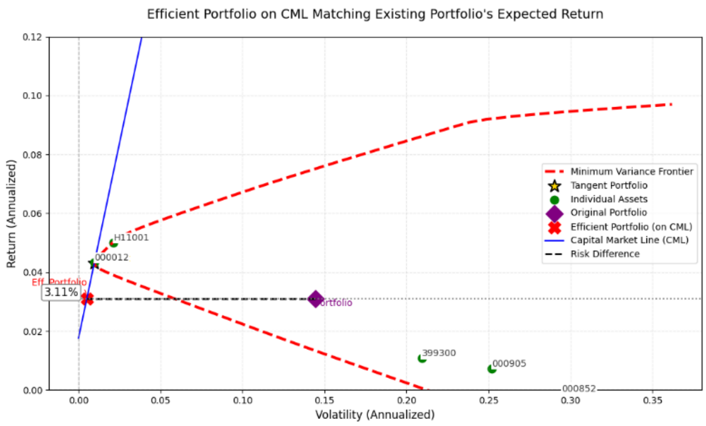

Figure 3: Efficient Portfolio on CML Matching Existing Portfolio's Expected Return

3.2 Efficient Portfolio with a Floor Return of 8%

This section explores the construction of an efficient portfolio that achieves a specific floor rate of return of 8% per year, again with the assumption that investment in a risk-free asset is possible.

The risk-free rate, return on tangency portfolio, and standard deviation of tangency portfolio are set at 1.76%, 4.31%, and 0.96% respectively. Plugging these values and the floor return of 8% into the following equations gives the required proportions and standard deviation.

$$w_{\text{tangency}} = \frac{R_{\text{floor}} - R_{\text{risk-free}}}{R_{\text{tangency}} - R_{\text{risk-free}}}$$

$$\sigma_{\text{floor}} = w_{\text{tangency}} \times \sigma_{\text{tangency}}$$

To achieve a target return of 8%, the investment proportions are as follows:

- **Weight of Risk-Free Asset:-1.4458**
- **Weight of Tangency Portfolio:2.4458**

The negative weight on the risk-free asset indicates that the fund would need to borrow at the risk-free rate to leverage its investment in the tangency portfolio. The expected standard deviation of returns for this portfolio, designed to meet the 8% floor return objective, is 2.3554%.

The detailed allocation of each asset to achieve an 8% floor rate is shown in the below table.

Asset	000012	000852	000905	399300	H11001	Risk-Free Asset
Opt. Prop.	243.0908%	0.0000%	0.0000%	1.4891%	0.0000%	-144.5799%

Table 5: Optimal Proportion of the Efficient Portfolio for 8% Floor Return

Figure 4 visually represents this efficient portfolio. The "Efficient Portfolio (floor 8%)" point on the CML clearly shows the combination of risk and return required to achieve the 8% floor return. This demonstrates how a higher expected return target compared to the existing portfolio's expected return necessitates a higher allocation to the risky tangency portfolio and, consequently, a higher expected standard deviation, yet it remains the most efficient way to achieve that return when a risk-free asset is available.

Figure 4: Efficient Portfolio on CML for 8% Floor Return

4 Incorporating Additional Risk Assets

4.1 Evaluation of Expanded Portfolio

We are interested in knowing if our portfolio should diversify its holdings by including real estate (represented by the Hang Seng REIT Index), HKEX listed stocks (represented by the Hang Seng Index), international stocks (represented by the S&P 500 Index), and commodities (represented by Gold ETF) in its portfolio.

4.1.1 Portfolio Frontier with Expanded Assets

We plot the portfolio frontier given the nine risky assets in which we are investing. Short sales are allowed when developing this frontier.

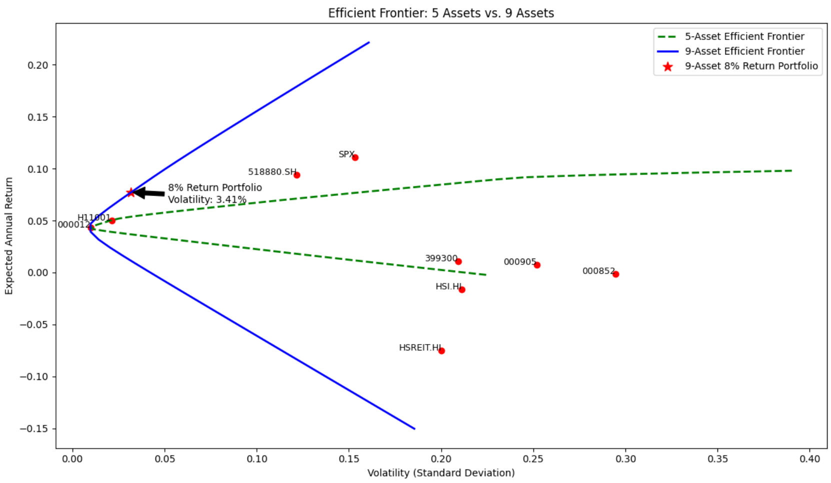

Figure 5: Plot of the efficient frontier with expanded assets

4.1.2 Efficient Portfolio Allocation

The efficient portfolio allocations required to achieve the “floor” rate of return of 8% per year with the expanded universe of assets are shown in the below table.

The optimal proportions of the expanded portfolio are shown in the below table.

Asset	000012	000852	000905	399300	H11001	HSREIT	518880	HSI	SPX
Opt. Prop.	17.44%	-1.47%	-1.72%	1.66%	69.00%	-14.39%	9.16%	3.20%	17.11%

Table 6: Optimal Proportion of the Expanded Portfolio

4.1.3 Expected Standard Deviation

The expected standard deviation of the portfolio with the expanded universe of assets, achieving the “floor” rate of return of 8% per year, is calculated to be 3.41%.

4.1.4 Reasons Supporting the Inclusion

1. Enhanced Risk-Return Profile: The 9-asset efficient frontier offers higher returns for the same level of risk compared with the 5-asset frontier.

2. Improved Diversification: The inclusion of alternative assets provides exposure to asset classes with low correlation to traditional Chinese equity indices (000012, 000852, etc.), as evidenced by the portfolio frontier’s outward shift.

3. Achievable Floor Return: For the required 8% return floor, the optimized 9-asset portfolio demonstrates superior efficiency. Our calculations show this portfolio can achieve the target return with an expected standard deviation of approximately 3.41%, compared to the 5-asset portfolio’s higher volatility of 17.30% for the same return.

4.2 Portfolio Allocation Under Constraints

Under the two constraints: a. You cannot take any short positions in risky assets. b. You cannot invest more than 30% of the total risky investment in any one asset category, the optimal allocation with an 8% annual return is shown in Table 7:

Asset	000012	000852	000905	399300	H11001	HSREIT	518880	HSI	SPX
Opt. Prop.	11.14%	0.00%	0.00%	0.00%	30.00%	0.00%	30.00%	0.00%	28.86%

Table 7: Optimal Proportion of the Expanded Portfolio Under Constraints

4.2.1 Expected Standard Deviation Under Constraints

The expected standard deviation of the portfolio under the given constraints, achieving the "floor" rate of return of 8% per year, is calculated to be 5.53%.

When investment constraints are imposed, some asset points may appear to the left of the efficient frontier. This is because these restrictions limit risk-reduction potential, leaving certain assets in suboptimal positions despite their individual risk-return characteristics.

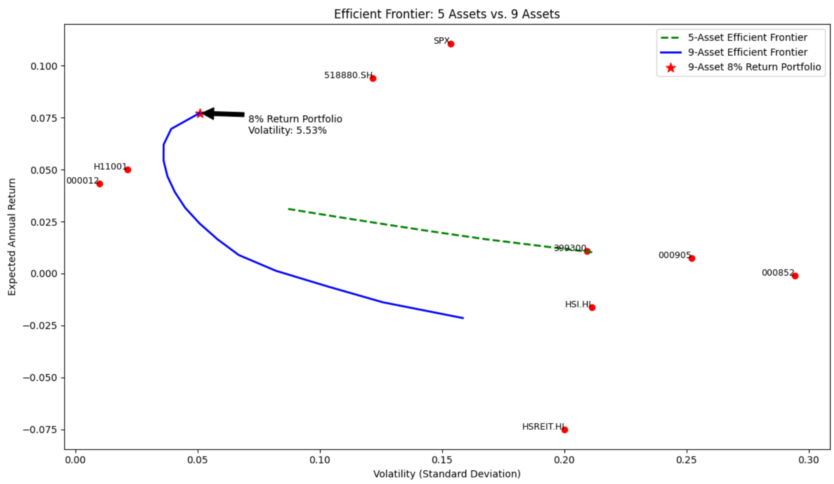

Figure 6: Plot of the efficient frontier under constraints

5 Argument for Asset Allocation Methods

To address the skepticism of board members who believe individual stock selection is more important than asset allocation, we present the following key arguments based on our research results and empirical evidence:

5.1 Diversification Benefits and Risk Reduction

Asset allocation provides inherent diversification, reducing unsystematic risk that cannot be eliminated through stock selection alone. Our analysis demonstrates that expanding the portfolio from 5 to 9 assets lowered the standard deviation for an 8% return target from 17.30% to 3.41%. Even under constraints (no short sales, 30% cap per asset), the 9-asset portfolio achieved a volatility of only 5.53%. Such risk mitigation is unattainable by concentrating on individual stocks.

5.2 Consistency with Modern Portfolio Theory (MPT)

Modern Portfolio Theory (MPT) emphasizes that asset allocation drives over 90% of portfolio performance variability. Our efficient frontier analysis confirms that optimizing asset weights—not stock picking—determines the risk-return trade-off. For instance, the tangency portfolio achieved a Sharpe ratio of 2.65, far surpassing the original portfolio's 0.094, solely through strategic allocation.

5.3 Cost and Operational Efficiency

Active stock selection incurs higher transaction costs, research expenses, and managerial effort. In contrast, asset allocation leverages passive indices, which are cost-effective and scalable. Our proposed 9-asset mix requires minimal rebalancing while capturing global market exposure.

6 Conclusion

This project underscores the superiority of asset allocation in achieving the fund's objectives. By rigorously evaluating performance statistics, plotting efficient frontiers, and optimizing portfolios under varying conditions, we demonstrated that:

- Diversification across asset classes (stocks, bonds, REITs, commodities) enhances returns while mitigating risk.
- Strategic allocation outperforms ad-hoc stock selection, as evidenced by Sharpe ratios and volatility metrics.
- The methodology is robust to real-world constraints, ensuring practical applicability.

For GBF College Fund, adopting the proposed 9-asset allocation—with disciplined rebalancing and adherence to MPT principles—will provide sustainable, risk-controlled growth. Stock selection, while valuable, cannot replicate these systemic advantages. Thus, we urge the board to prioritize asset allocation as the cornerstone of the fund's investment policy.